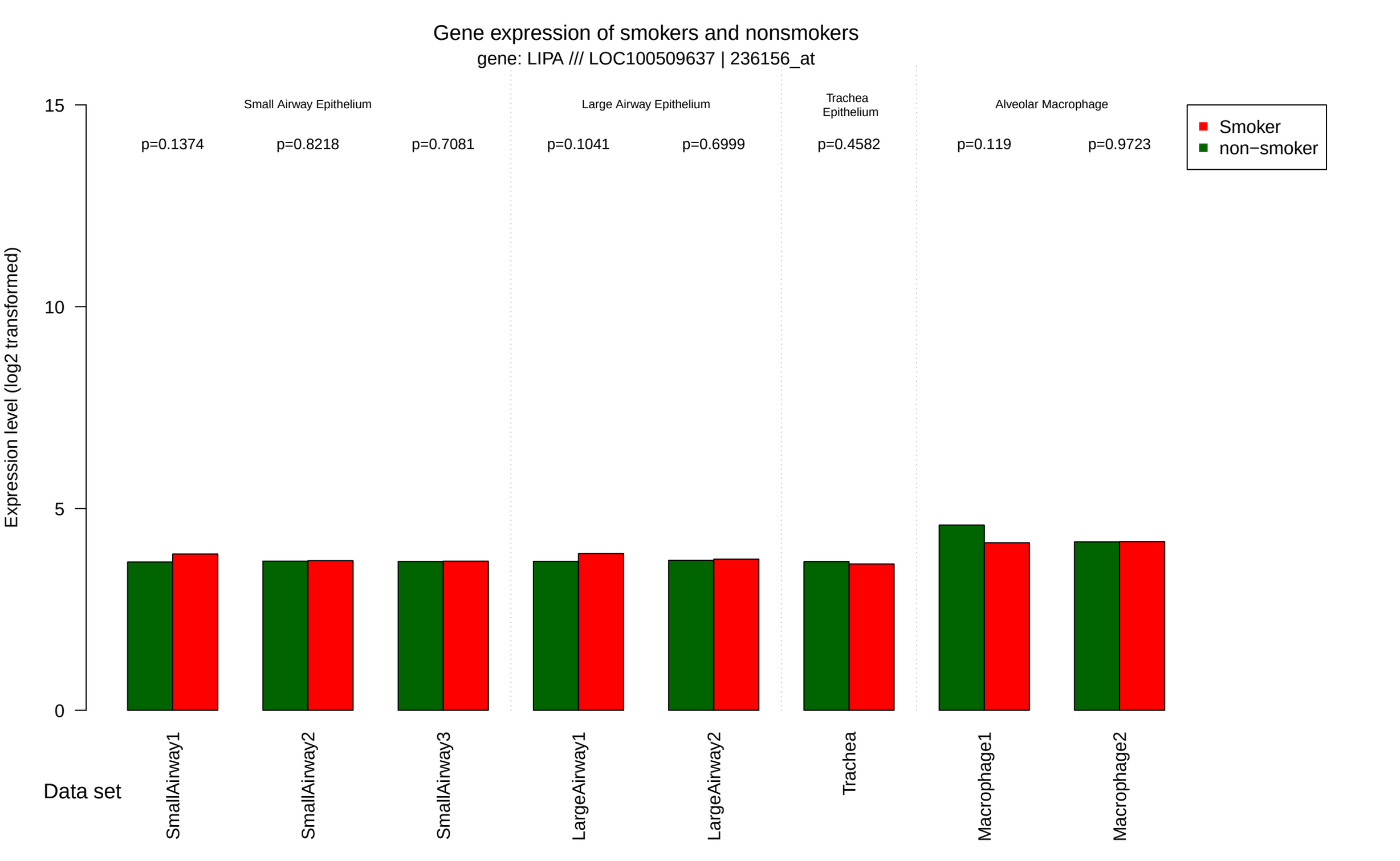
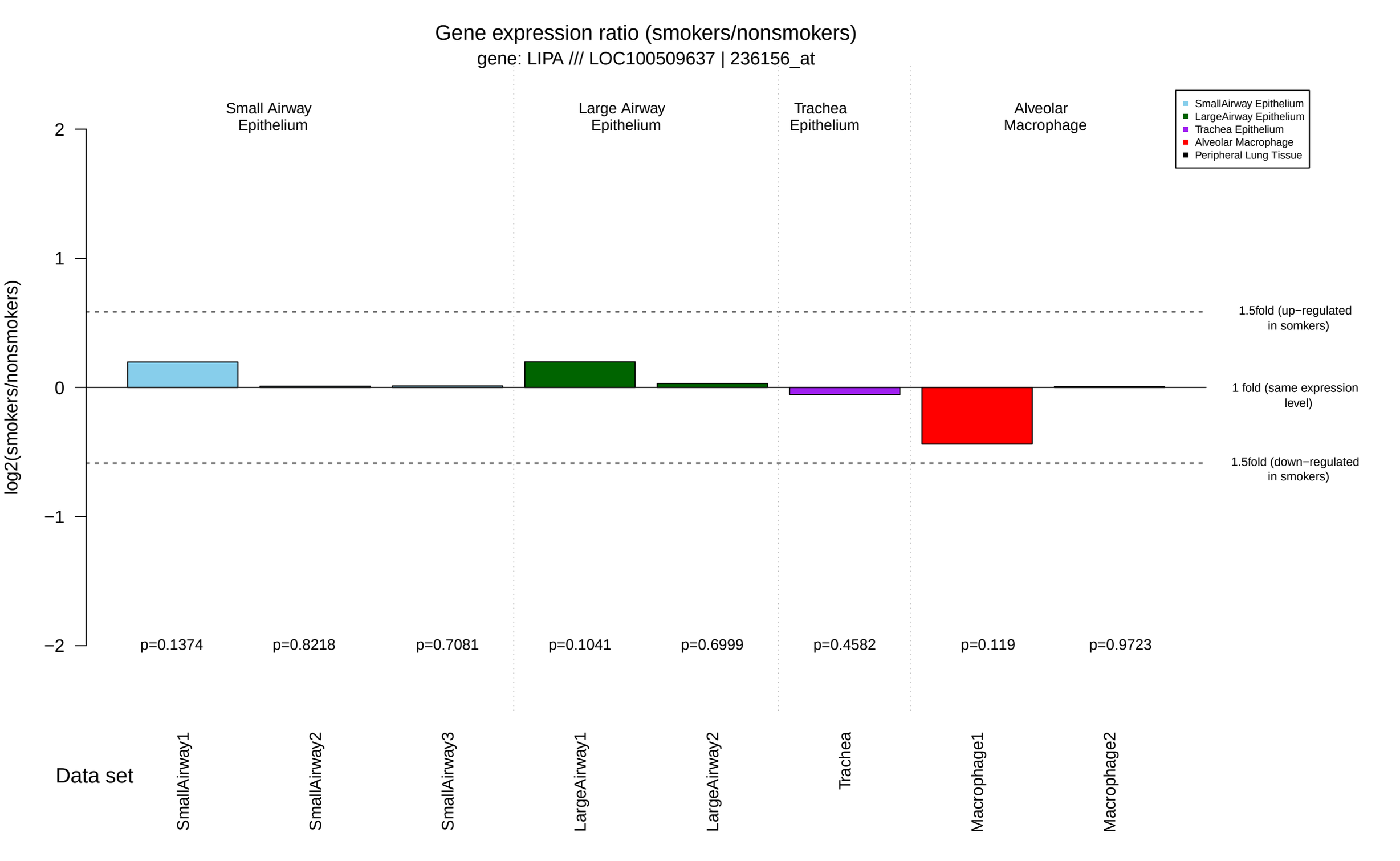
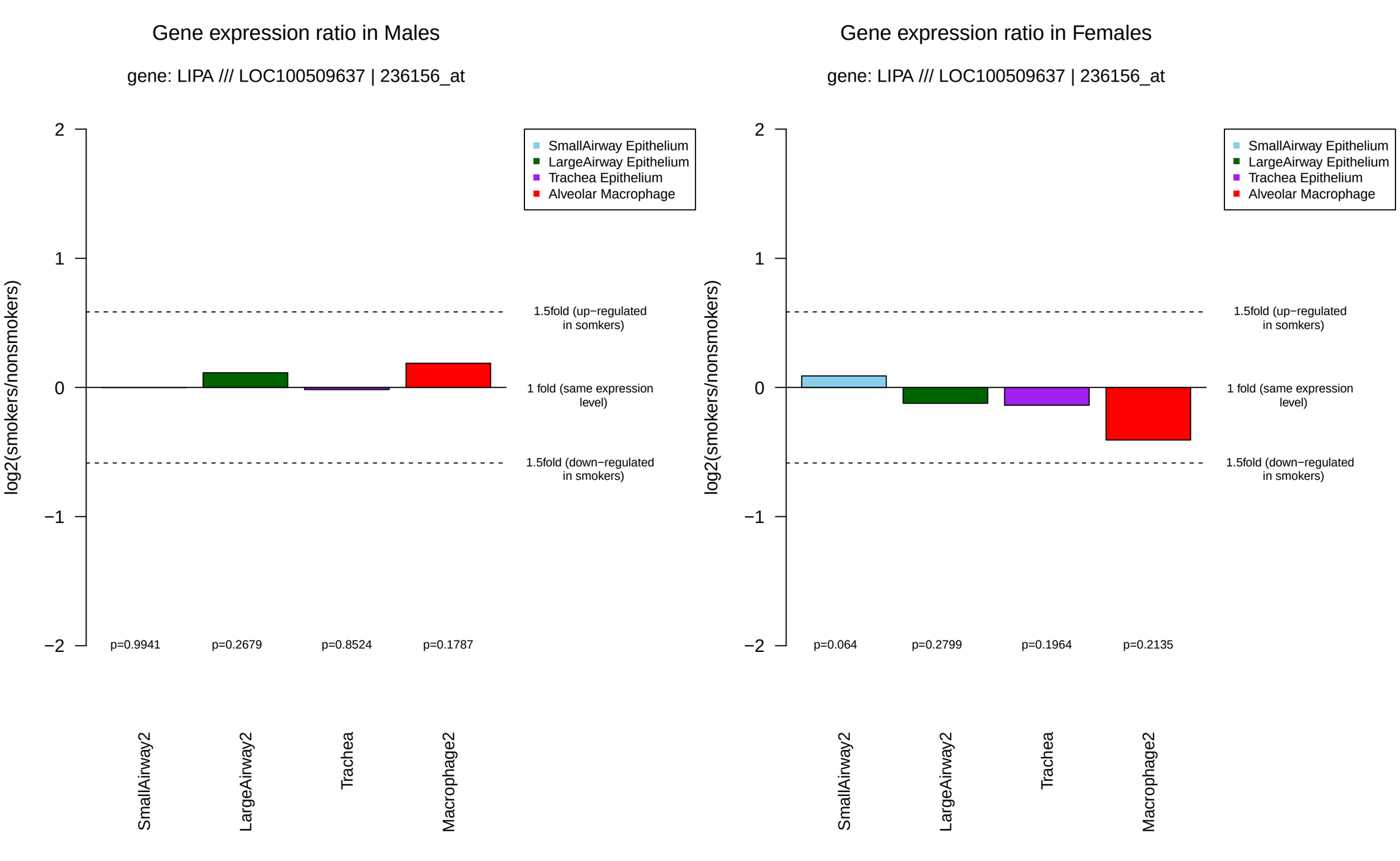
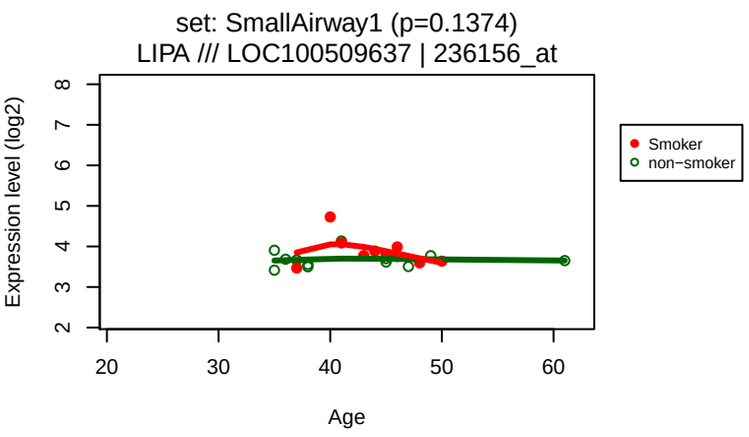
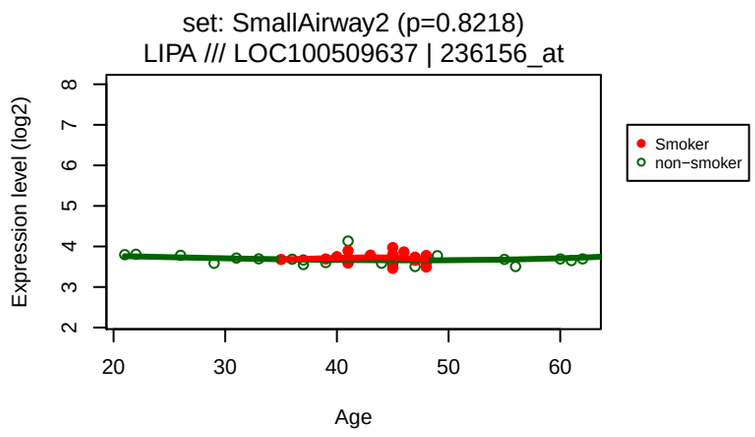
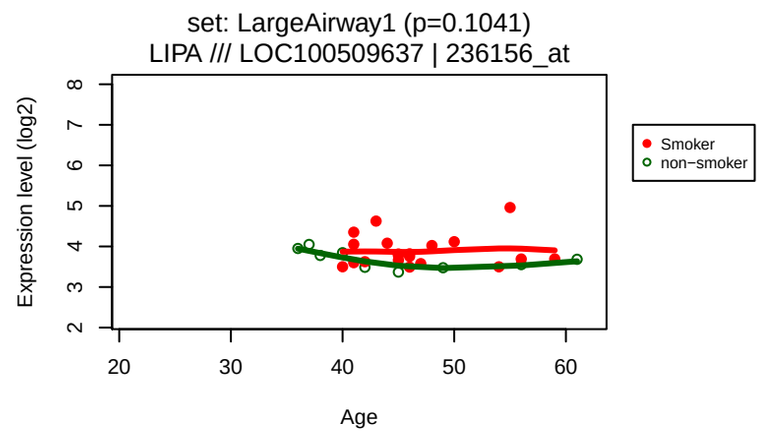

Gene expression of smokers and nonsmokers
gene: LIPA /// LOC100509637 | 236156_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1374
p=0.8218
p=0.7081
p=0.1041
p=0.6999
p=0.4582
p=0.119
p=0.9723
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

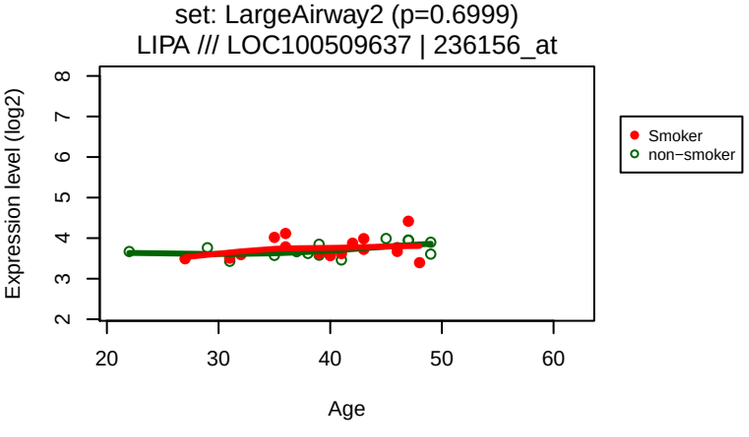

Gene expression ratio (smokers/nonsmokers)
gene: LIPA /// LOC100509637 | 236156_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1374
p=0.8218
p=0.7081
p=0.1041
p=0.6999
p=0.4582
p=0.119
p=0.9723
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

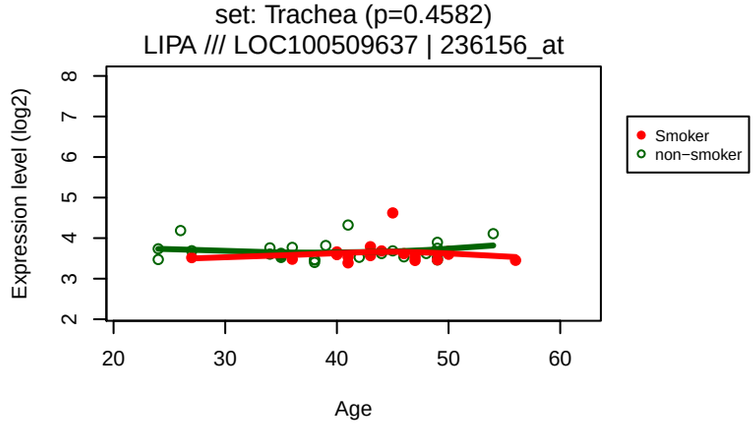

Gene expression ratio in Males
gene: LIPA /// LOC100509637 | 236156_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9941
p=0.2679
p=0.8524
p=0.1787
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: LIPA /// LOC100509637 | 236156_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.064
p=0.2799
p=0.1964
p=0.2135
SmallAirway2
LargeAirway2
Trachea
Macrophage2

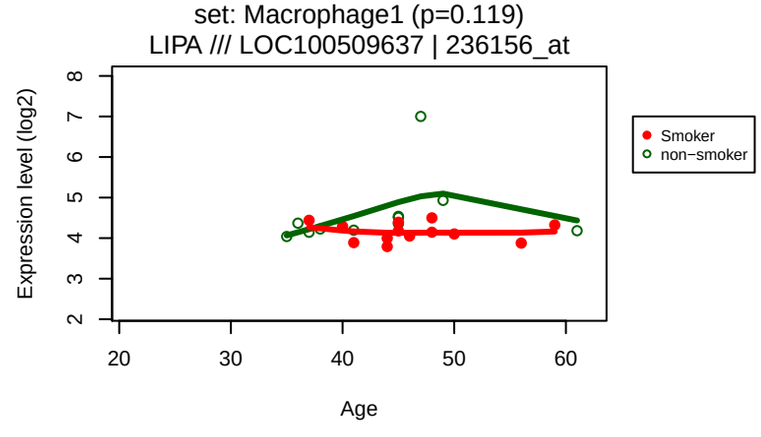

set: SmallAirway1 (p=0.1374)
LIPA /// LOC100509637 | 236156_at
Expression level (log2)
Age
Smoker
non−smoker

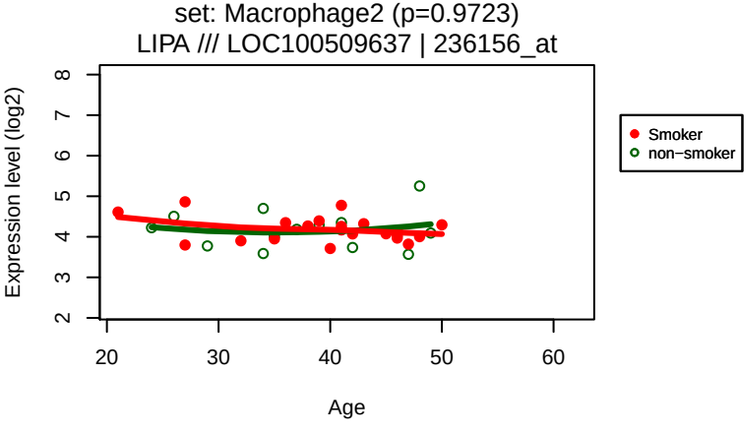

set: SmallAirway2 (p=0.8218)
LIPA /// LOC100509637 | 236156_at
Expression level (log2)
Age
Smoker
non−smoker

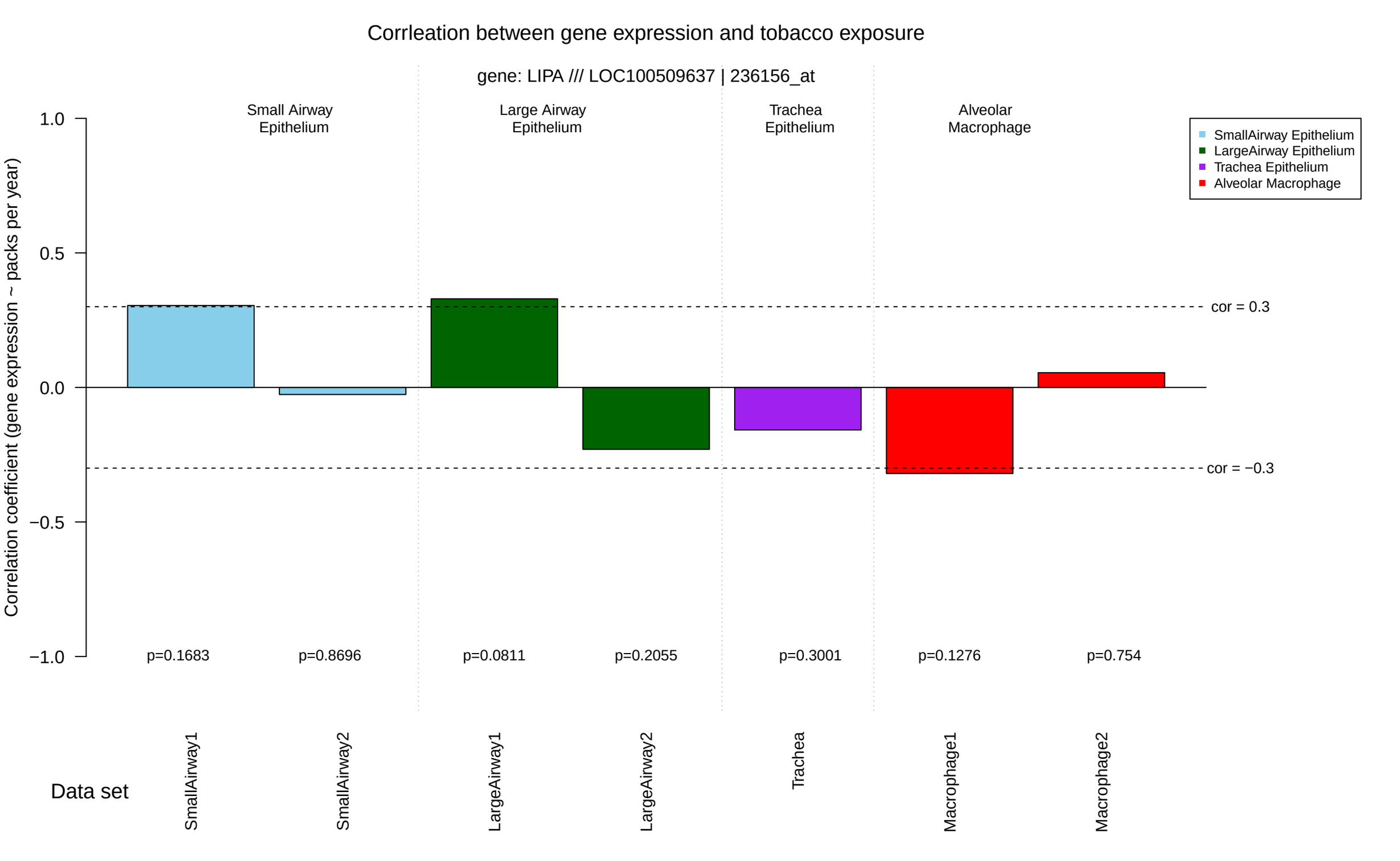

set: LargeAirway1 (p=0.1041)
LIPA /// LOC100509637 | 236156_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6999)
LIPA /// LOC100509637 | 236156_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4582)
LIPA /// LOC100509637 | 236156_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.119)
LIPA /// LOC100509637 | 236156_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9723)
LIPA /// LOC100509637 | 236156_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: LIPA /// LOC100509637 | 236156_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1683
p=0.8696
p=0.0811
p=0.2055
p=0.3001
p=0.1276
p=0.754
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2